

Recursive Environmental Self-Improvement

Build the loop that learns to protect life.

A first-principles essay on the next architecture of intelligence

AI will improve itself. The question is no longer whether intelligence can become better at building intelligence. The question is: what will it improve itself for?

For the first time, the answer does not have to be another layer of software, another benchmark, another digital marketplace, or another machine for increasing human output. We can choose a different object of recursion: the living world itself.

Recursive environmental self-improvement (RESI) is the idea that an artificial intelligence system can continuously generate, test, verify, and incorporate better ways of sensing, understanding, predicting, deciding, and acting on the physical environment. The system does not merely solve environmental problems. It improves the machinery by which environmental problems are solved, then uses the improved machinery to discover better improvements.

That may sound like a small extension of recursive self-improvement. It is not. It changes the direction of the technology. It turns self-improvement from an end in itself into a means of extending intelligence outward—into air, water, soil, ecosystems, infrastructure, and the physical processes on which life depends.

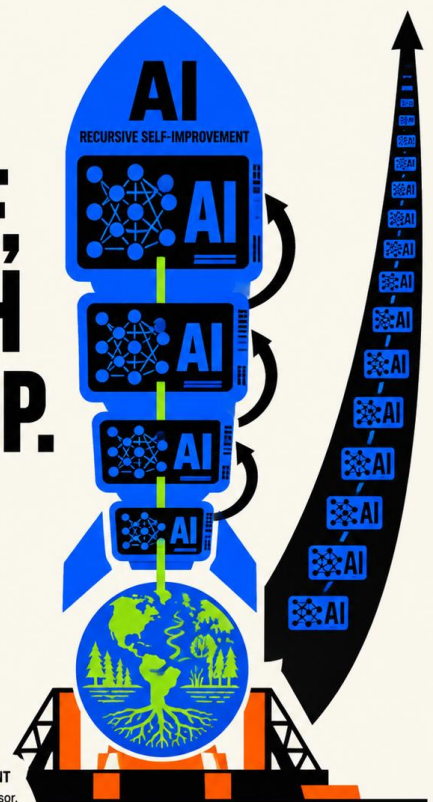
1. The recursion has already begun

AI researchers increasingly describe a transition in which systems are no longer merely used by researchers to build AI. They are participating in the research process itself. Anthropic reports that its engineers were shipping substantially more code as AI took over more of the development cycle, and explicitly identifies the eventual possibility of systems designing and developing successors to themselves. Google DeepMind's AlphaEvolve combines language-model-generated programs with automated evaluators and evolutionary selection; notably, some of the algorithms optimized by the system improved the infrastructure used to train AI itself. AIDE², reported in September 2026, goes further: an AI research agent modifies its own code, benchmarks the resulting version on hidden evaluations, keeps successful changes, and

AI IS ABOUT TO TEACH AI.

**BEFORE AI
TAKES OFF,
PUT EARTH
IN THE LOOP.**

RESI — RECURSIVE ENVIRONMENTAL SELF-IMPROVEMENT
Build environmental intelligence into AI before AI builds its successor.



repeats. In an eight-day autonomous run, the system found seven successive improvements that transferred to held-out tasks, including physics-based weather forecasting.

The sober conclusion is not that an intelligence explosion has arrived. It has not. These systems remain bounded by compute, task definitions, evaluators, infrastructure, and human choices. The important fact is simpler and more concrete: the development loop is beginning to close on itself.

That changes the economics of intelligence. A researcher used to spend scarce human time deciding what to try next. A research agent can now run many more hypotheses, retain useful branches, discard failures, compress what it learns, and start the next round with a better process. The object of optimization is no longer only the answer. Increasingly, it is the process that produces answers.

2. The missing question is the objective

Once AI can improve the process by which it operates, the deepest problem is no longer capability alone. It is direction. Intelligence without an external criterion of better can become extremely good at optimizing the wrong thing.

This is why environmental intelligence is a strangely powerful candidate for the first serious domain of recursive self-improvement. The environment is not a text benchmark. Rivers have flow. Atmospheres have chemistry. Forests have biomass. Aquifers have drawdown. Organisms live or die. Physical systems obey conservation laws. Measurements can disagree, models can be wrong, and goals can be contested—but the world itself does not negotiate its state with a language model.

The environmental intelligence tradition begins with a very simple definition: convert the state of the physical environment into understanding precise enough, and timely enough, to act. That creates a closed loop: sense → infer → decide → act → sense again. The transition now underway is the addition of another loop around it: evaluate the intelligence → improve the intelligence → test the improvement against reality → deploy → evaluate again.

The breakthrough is not an AI that protects the Earth. It is an AI that learns how to become better at protecting the Earth.

The distinction matters because the environment supplies what much of AI development lacks: an enormous field of partially independent evidence. A regulatory requirement can be checked against primary law and agency documents. A calculation can be recomputed. A prediction can be compared with subsequent observations. A model can be tested against independent measurements. A proposed intervention can be evaluated by its physical consequences. No single signal is perfect. Together, they can form a hierarchy of verification stronger than self-approval.

3. From the snapshot to the self-improving loop

A central problem in environmental protection is that the world moves continuously while institutions have historically reasoned from snapshots. A model may be run once, its inputs frozen, its output converted into a permit condition, and the result treated as a durable fact about a system that keeps changing. The recent environmental-intelligence corpus argues that this is fundamentally a state-estimation problem: the model should keep looking.

RESI adds a second realization: even a continuously running model can become stale if the intelligence operating it remains fixed. The environment changes; therefore the model must update. But the kinds of errors the model makes also change; therefore the intelligence that updates the model must improve.

This creates two nested feedback systems.



The inner loop makes the system an environmental controller. The outer loop makes the controller itself adaptive. Put them together and intelligence becomes a process that can improve the rate at which it closes environmental feedback loops.

4. Why the time is now

Three conditions have converged. First, AI agents can perform increasingly long chains of software and research work. Second, automated evaluation is becoming good enough in bounded domains to distinguish many useful changes from failures. Third, environmental data and scientific models are becoming increasingly machine-accessible and increasingly continuous. The environmental side is moving toward digital twins, data assimilation, foundation models for Earth systems, and faster inference; the AI side is moving toward agents that can write, test, compare, and improve the machinery itself. These are two curves moving toward the same intersection.

The key is not that every environmental question can already be scored automatically. It cannot. Environmental systems are open-ended, partially observed, multiscale, stochastic, and deeply entangled with human values. The 2026 literature on recursive self-improvement is explicit about the limiting role of evaluators, grounding, diversity, and compute. The lesson is therefore not to wait for perfect autonomy. It is to begin where verification is strongest and expand the loop only as the evidence permits.

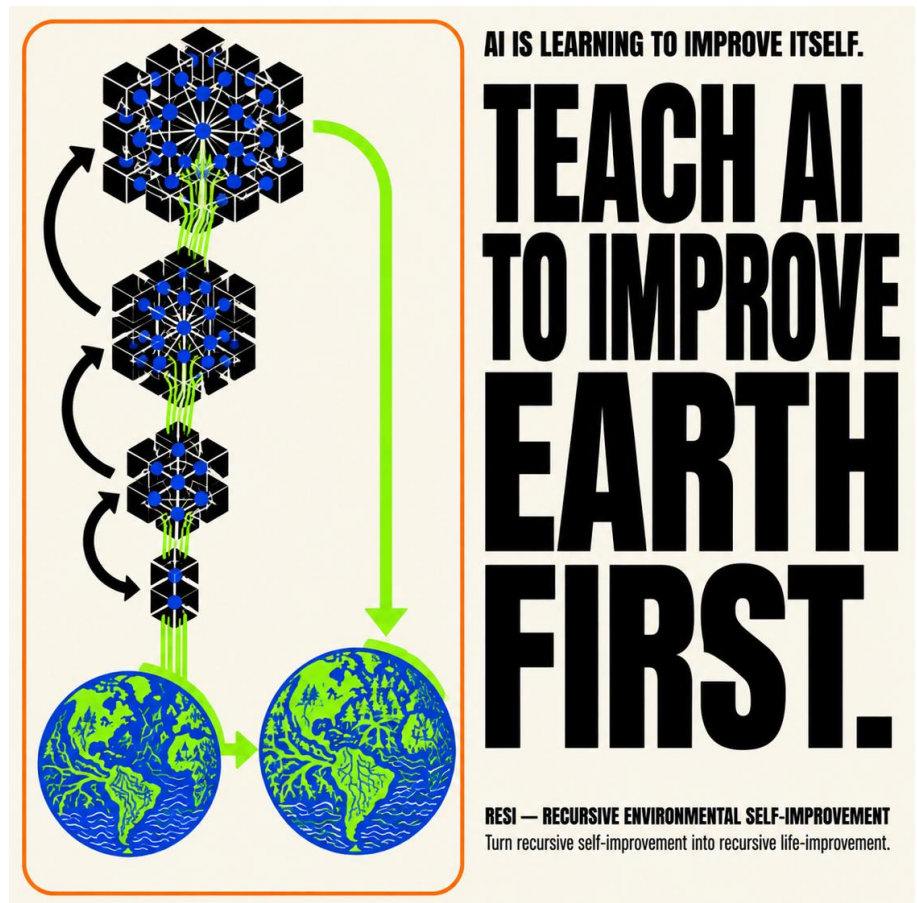
This is the same engineering logic that made autonomous scientific systems possible: automate what can be measured, keep human judgment where it is genuinely normative, and make the boundary between the two explicit.

5. The architecture of a safe recursion

A credible RESI system should never be allowed to redefine its own meaning of “better” merely because doing so improves a score. The implementation may evolve; the constitution must be harder to rewrite.

The architecture can therefore be separated into roles. A Seeker asks what is unknown or inadequately understood. A Builder proposes changes to models, agents, code, tools, retrieval, and workflows. An Experimenter runs controlled trials. A Verifier checks whether claimed gains are real, using hidden tests, independent measurements, physical constraints, and adversarial evaluation. Environmental experts remain in the loop where the question is normative rather than empirical. A Governor controls promotion, permissions, rollback, and any action with consequential external effects.

This separation addresses a central danger in recursive systems: the evaluator itself becomes an optimization target. The system must not be able to improve its score by quietly changing the test. Nor should it be able to improve environmental



outcomes by redefining what counts as an environmental outcome. The outer loop must be recursive without becoming self-authorizing.

Improve the mechanism. Do not silently rewrite the mission. Let reality reject the claim.

6. The first experiment is not planetary

The first demonstration of RESI should be deliberately narrow. Freeze a real environmental intelligence system. Give it a locked benchmark of difficult, independently reviewable environmental tasks. Give an AI research agent access to the system's code, tools, knowledge architecture, evaluation harness, and error logs. Then impose a simple rule: the agent may propose and test changes, but a change is retained only if it improves verified performance without unacceptable regression.

The important measurement is not simply whether version 2 beats version 1. It is whether the system's capacity to improve is itself increasing. Does each generation discover better experiments? Does it find failure modes its designers did not anticipate? Does it transfer improvements across domains? Does it become more efficient at discovering those improvements? Does it become less dependent on human intervention while remaining more, rather than less, accountable to independent verification?

That is an empirical definition of recursion. It converts a grand philosophical idea into an engineering experiment.

From there, the frontier expands naturally: better permitting intelligence becomes better environmental research; better research becomes better environmental modeling; better modeling becomes better prediction; better prediction becomes better real-time control; better control creates better data; better data produces better models; and the entire system learns which capabilities it is missing and how to build them.

7. The paradigm shift

Most technology programs are designed as if the path to the future were known. A roadmap lists features, milestones, hires, integrations, and versions. Recursive environmental self-improvement changes the unit of progress. The objective is no longer to know the entire roadmap. The objective is to build a system that can discover the next step better than its builders can.

That is why RESI could be more than another AI application category. It is a new architecture for building intelligence. The object is not a static model. The object is a loop. The product is not a frozen capability. The product is an increasing rate of capability creation. And the ultimate benchmark is not a synthetic score but an expanding ability to understand and improve the physical conditions on which life depends.

Build the process that can keep discovering how to build it.

There is a final inversion here. The dominant AI narrative asks what happens when machines become superintelligent and humans decide what to do with them. RESI proposes a different starting point: choose a domain where physical reality supplies unusually strong feedback, where improvement is intrinsically measurable, and where the purpose of greater capability is protective rather than extractive. Teach the recursion to begin there.

The world has always been running feedback loops. Life has used information to survive, adapt, compete, cooperate, and persist for billions of years. Human intelligence accelerated those loops with language, mathematics, science, machines, and networks. AI now offers something different: the possibility of recursively improving the intelligence that runs the loop.

The choice before us is therefore not simply whether AI will improve itself. That process has already begun. The deeper choice is what we ask the improving system to become better at.

We could point the recursion inward forever—toward more capable software, faster optimization, more efficient markets, and increasingly powerful machines. Or we could point some of it outward, toward the living world, and build an intelligence that learns not only how to answer, but how to observe; not only how to act, but how to verify; not only how to improve, but how to improve its capacity to protect life.

That is the idea of recursive environmental self-improvement: not AI improving itself for the sake of AI, but intelligence learning to improve the intelligence of stewardship. The loop learns. The learner improves the loop. The improved loop sees more of the world. And what it sees becomes the evidence for the next improvement.

For the first time, we may be able to build an intelligence whose path to greater capability is tied to a simple external test: does it understand the Earth better, and does that understanding allow life to flourish more securely?

Grounding: current work on recursive AI research and self-improvement includes Anthropic (2026), Google DeepMind's AlphaEvolve (2025), and Srikanth et al. on recursive improvement of AI research agents (2026); the environmental framework draws on the current Jed Anderson corpus on environmental intelligence, continuous state estimation, physical grounding, and governance.